

• Hilbert Spaces. Chapter 6.

• Inner product (scalar product). X v.sp. $(\cdot, \cdot): X \times X \rightarrow \mathbb{C}$ or $\mathbb{R}$

such that: (a) $(x, \lambda y + \mu z) = \lambda (x, y) + \mu (x, z)$ (linearity)

$$(b) (x, y) = \overline{(y, x)}$$

$$(c) (x, x) \geq 0$$

$$(d) (x, x) = 0 \Rightarrow x = 0 \quad (\text{positive definite}).$$

• $(X, (\cdot, \cdot))$ is called pre-Hilbert space.

$$(b) (\lambda x + \mu y, z) = \overline{(z, \lambda x + \mu y)} = \overline{\lambda (z, x) + \mu (z, y)} = \overline{\lambda} (x, z) + \overline{\mu} (y, z) \quad (\text{anti-linearity})$$

• When X is real v.sp., then $(\cdot, \cdot) \rightarrow \mathbb{R}$ is bilinear form.

• complex v.s.p., $\mathbb{C}$ is sesquilinear form.

• $\|x\| = (x, x)^{1/2}$ defines a norm (as we will verify shortly).

• $(X, (\cdot, \cdot))$ is therefore a normed vector space with that norm.

Def. A Hilbert space is a complete pre-Hilbert space.

Examples: $\mathbb{C}^n$ $(x, y) = \sum_{i=1}^n \overline{x_i} y_i$

$\mathcal{C}([a, b])$ $(f, g) = \int_a^b \overline{f(x)} g(x) dx$.

$(\mathcal{C}, (\cdot, \cdot))$ is pre-Hilbert. Completion is $L^2(a, b)$.

$\mathcal{C}^k([a, b])$ $(f, g)_k = \sum_{j=0}^k \int_a^b \overline{f^{(j)}(x)} g^{(j)}(x) dx$.

$(\mathcal{C}^k, (\cdot, \cdot)_k)$ is pre-Hilbert. Completion is $H^k(a, b) \equiv W^{k,2}(a, b)$
(Hilbert or Sobolev space).

• Theorem Cauchy-Schwarz. (x, y) in X . Then $|(x, y)| \leq \|x\| \|y\|$.

• proof. $(\cdot, \cdot) \rightarrow \mathbb{R}$. $\lambda \in \mathbb{R}$. $0 \leq \|x - \lambda y\|^2 = (x - \lambda y, x - \lambda y) = \|x\|^2 + \lambda^2 \|y\|^2 - 2\lambda (x, y)$.

• So, using $\pm \lambda$, we find $2|\lambda| |(x, y)| \leq \|x\|^2 + \lambda^2 \|y\|^2$.

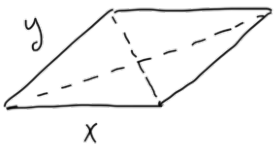
• above clear for $y=0$. For $y \neq 0$, take $\lambda = \frac{\|x\|}{\|y\|}$ to get $2|(x, y)| \leq 2\|x\| \|y\|$ $\square$
or $x=0$ $x \neq 0$

• Corollary $\|x+y\| \leq \|x\| + \|y\|$ so $\|\cdot\|$ is indeed a norm.

proof: $\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2(x, y) \leq \|x\|^2 + \|y\|^2 + 2\|x\| \|y\| = (\|x\| + \|y\|)^2$ $\square$

. Theorem Parallelogram Law. $(X, \|\cdot\|)$ is an inner product space

iff for all x, y in X , $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$.



. proof $\Rightarrow$ is clear check

$\Leftarrow$ define $(x, y) = \frac{1}{4} (\|x+y\|^2 - \|x-y\|^2)$ and verify that this is indeed an inner product (not so easy!).

. product space on $X \times Y$, define: $((x_1, y_1), (x_2, y_2))_{X \times Y} := (x_1, x_2)_X + (y_1, y_2)_Y$

Then $(X \times Y, (\cdot, \cdot)_{X \times Y})$ pre-Hilbert.

. Theorem $(\cdot, \cdot) : X \times X \rightarrow \mathbb{C}$ is continuous (for $(X, \|\cdot\|)$)

Proof: $|(x_1, y_1) - (x_2, y_2)| = |(x_1 - x_2, y_1) - (x_2, y_2 - y_1)|$

C.S.

$\leq \|x_1 - x_2\| \|y_1\| + \|x_2\| \|y_2 - y_1\|$ so continuous.

• Orthogonality. $\mathcal{H}$: Hilbert space.

• $x \perp y$ if $(x, y) = 0$ x orthogonal to y ; y orthogonal to x .

• $A \perp B$ if $x \perp y$ for all $x \in A$ and $y \in B$.

• $A^\perp = \{ x \in \mathcal{H}, x \perp y, y \in A \}$ orthogonal complement of A .

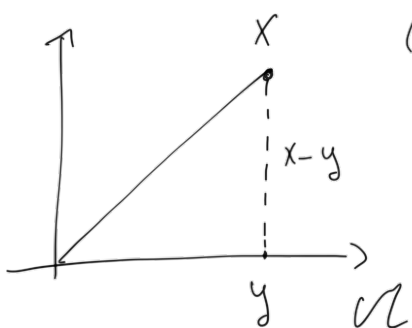
Theorem $A^\perp$ is a closed linear subspace of $\mathcal{H}$. . proof: check.

• Theorem. Projection onto $\mathcal{M}$ a closed linear subspace of $\mathcal{H}$.

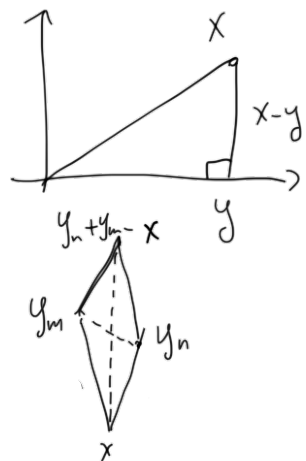
(a) $\forall x \in \mathcal{H}, \exists ! y \in \mathcal{M}$ s.t. $\|x - y\| = \min_{z \in \mathcal{M}} \|x - z\|$.

(b) $y \in \mathcal{M}$, closest point to x in $\mathcal{M}$, is unique point such that

$$x - y \in \mathcal{M}^\perp.$$



• proof of (a) unique minimizer of $\|x-z\|$ for $z \in \mathcal{M}$.



• Let $x \in \mathcal{H}$ and define $d = \inf \{ \|x-z\|, z \in \mathcal{M} \}$. Then there is a minimizing sequence $(y_n)_{n \geq 1}$ s.t. $\|x-y_n\| \leq d + \frac{1}{n}$.

parallelogram law $\Rightarrow \|y_n - y_m\|^2 + 4\|x - \frac{y_n + y_m}{2}\|^2 = 2\|x - y_n\|^2 + 2\|x - y_m\|^2$.

$$\text{So, } \|y_n - y_m\|^2 \leq 2\left(d + \frac{1}{n}\right)^2 + 2\left(d + \frac{1}{m}\right)^2 - 4d^2 \text{ since } \|x - \frac{y_n + y_m}{2}\| \geq d.$$

$$\leq 4d\left(\frac{1}{n} + \frac{1}{m}\right) + 2\left(\frac{1}{n^2} + \frac{1}{m^2}\right) \rightarrow 0 \text{ as } n, m \rightarrow \infty.$$

• Thus, (y_n) Cauchy $y_n \rightarrow y \in \mathcal{M}$ since $\mathcal{M}$ closed. $d = \|x - y\|$.

Uniqueness: $0 \leq \|y - \tilde{y}\|^2 = 2\|x - \tilde{y}\|^2 + 2\|x - y\|^2 - 4\|x - \frac{y + \tilde{y}}{2}\|^2 \leq 4d^2 - 4d^2 = 0 \square$

(b) Let y be as above and $z \in \mathcal{M}$. $y - \lambda z \in \mathcal{M}$. We want $(x-y, z) = 0 \forall z \in \mathcal{M}$.

$$\text{From (a), } \|x-y\|^2 \leq \|x-y + \lambda z\|^2 \Rightarrow 0 \leq 2(\pm \lambda)(x-y, z) + \lambda^2 \|z\|^2$$

$$\text{Thus, } 2|\lambda|(x-y, z) \leq \lambda^2 \|z\|^2. \text{ Send } \lambda \downarrow 0 \text{ to get } (x-y, z) = 0.$$

It remains to prove uniqueness. Let $x-y \in \mathcal{M}^\perp$ and $x-\tilde{y} \in \mathcal{M}^\perp$, $y, \tilde{y} \in \mathcal{M}$.

Then $y - \tilde{y} \in \mathcal{M}$ and $(y - \tilde{y}, x-y) = 0 = (y - \tilde{y}, x-\tilde{y})$ by assumption.

$$\text{Difference } \Rightarrow \|y - \tilde{y}\|^2 = 0 \Rightarrow y = \tilde{y}.$$

□

- Let $\mathcal{M}$ and $\mathcal{N}$ be closed linear subspaces of $\mathcal{H}$ with $\mathcal{M} \perp \mathcal{N}$.
- Def. The direct sum $\mathcal{M} \oplus \mathcal{N} = \{ y + z ; y \in \mathcal{M}, z \in \mathcal{N} \}$.
- Corollary $\mathcal{M}$ closed linear subspace of $\mathcal{H}$. Then $\boxed{\mathcal{H} = \mathcal{M} \oplus \mathcal{M}^\perp}$.
- proof $x \in \mathcal{H}$, y as in projection theorem. Then $x = \underbrace{y}_{\in \mathcal{M}} + \underbrace{x-y}_{\in \mathcal{M}^\perp} \quad \square$
- If $\mathcal{M}$ not closed, then $\boxed{\mathcal{H} = \overline{\mathcal{M}} \oplus \mathcal{M}^\perp}$.
- Such a decomposition is not always available for Banach spaces.

Orthogonal bases.

- Def. • a set $U \subset H$ is orthogonal if $u \perp v$ for all $u, v \in U$ s.t. $u \neq v$.
 • an orthogonal set is orthonormal if $\|u\| = 1 \ \forall u \in U$.

• Let I be index set (countable or not). $J \subset I$ finite subset.

Def. unordered sum $S_J = \sum_{\alpha \in J} x_\alpha$, $x_\alpha \in H$.

Def. • An unordered sum converges unconditionally to $x = \sum_{\alpha \in I} x_\alpha$
 if $\forall \varepsilon > 0$, $\exists J^\varepsilon$ finite such that $\|S_J - x\| \leq \varepsilon$
 $\left. \begin{array}{l} \forall J \text{ such that:} \\ J^\varepsilon \subset J \text{ finite} \subset I. \end{array} \right\}$

• The convergence is then independent of the re-ordering of terms.

• $\sum_{\alpha \in I} x_\alpha$ converges absolutely if $\sum_{\alpha \in I} \|x_\alpha\|$ c.v. unconditionally.

• Absolute c.v. $\Rightarrow$ unconditional c.v. but not $\Leftarrow$!

example: $\sum_{n=1}^{\infty} \frac{1}{n} e_n$ $\{e_n\}$ orthonormal.

Indeed, $\left\| \sum_{n=1}^N \frac{1}{n} e_n \right\|^2 = \sum_{n=1}^N \frac{1}{n^2} < \infty$ while $\sum_{n=1}^N \frac{1}{n} \sim \ln N \nearrow \infty$.

• Lemma $U = \{u_\alpha, \alpha \in I\}$ orthogonal set in $\mathcal{H}$.

Then $\sum_{\alpha \in I} u_\alpha$ c.v. unconditionally $\Leftrightarrow \sum_{\alpha \in I} \|u_\alpha\|^2 < \infty$.

• Proof $\left\| \sum_{\alpha \in J} u_\alpha \right\|^2 = \left(\sum_{\alpha} u_\alpha, \sum_{\beta} u_\beta \right) = \sum_{\alpha \in J} \|u_\alpha\|^2$ so clear (check).

• Theorem Bessel's inequality. $U = \{u_\alpha, \alpha \in I\}$ orthonormal in $\mathcal{H}$.

$$x \in \mathcal{H}. \quad \left(\begin{array}{l} (a) \sum_{\alpha \in I} |(u_\alpha, x)|^2 \leq \|x\|^2 \end{array} \right.$$

$$\left(\begin{array}{l} (b) x_u = \sum_{\alpha \in I} (u_\alpha, x) u_\alpha \quad : \text{convergent sum} \end{array} \right.$$

$$\left(\begin{array}{l} (c) x - x_u \in U^\perp. \end{array} \right.$$

: from orthonormality; check.

• proof. $\left\| x - \sum_{\alpha \in J} (u_\alpha, x) u_\alpha \right\|^2 = \|x\|^2 - \sum_{\alpha \in J} |(u_\alpha, x)|^2 \geq 0 \Rightarrow (a)$

(b) Above lemma implies convergence. Call x_u the result.

$$(c) u_\beta \in U. \quad (x - x_u, u_\beta) = (x, u_\beta) - \left(\sum_{\alpha} (u_\alpha, x) u_\alpha, u_\beta \right)$$

$$= (x, u_\beta) - \sum_{\alpha} \overline{(u_\alpha, x)} (u_\alpha, u_\beta) = (x, u_\beta) - \overline{(u_\beta, x)} = 0 \quad \square$$

• Definition (Span) | $[U] = \left\{ \sum_{u \in U} c_u u \mid c_u \in \mathbb{C}, \sum_u |c_u|^2 < \infty \right\}$.
 U orthonormal | $[U]$ is smallest closed subspace of $\mathcal{H}$ containing U .

Theorem. $U = \{u_\alpha, \alpha \in I\}$ orthonormal set in $\mathcal{H}$.

The following statements are equivalent:

(a) $(u_\alpha, x) = 0 \quad \forall \alpha \in I \Rightarrow x = 0$.

(b) $x = \sum_{\alpha \in I} (u_\alpha, x) u_\alpha \quad \forall x \in \mathcal{H}$.

(c) $\|x\|^2 = \sum_{\alpha \in I} |(u_\alpha, x)|^2 \quad \forall x \in \mathcal{H}$. Parseval identity.

(d) $[U] = \mathcal{H}$.

• proof. $b \Leftrightarrow d$. $a \Rightarrow x = x_U$ in Bessel's inequality $\Leftrightarrow b \xRightarrow{\text{Lemma}} c \Rightarrow a$.

• Definition. U is complete if $[U] = \mathcal{H}$. Then U called orthonormal basis.

Theorem. If onb $x = \sum_{\alpha} a_{\alpha} u_{\alpha}$ $y = \sum_{\alpha} b_{\alpha} u_{\alpha}$

Then $(x, y) = \sum_{\alpha} \bar{a}_{\alpha} b_{\alpha}$. (check)

Theorem. Every Hilbert space admits an orthonormal basis

. If U is orthonormal set, then it is contained in a onb.

. proof Zorn's Lemma

Corollary. Assume H is separable (admits a countable basis.)

Then $H \cong \ell^2(\mathbb{N})$

. Indeed identifying basis elements generates an isomorphism.

. This identification is not always useful or enlightening.

But there is only one separable Hilbert space.

Remark. Banach-Mazur Theorem: Every real, separable Banach space $(X, \|\cdot\|)$ is isometrically isomorphic to a closed subspace of $C^0([0, 1]; \mathbb{R})$.

• Gram Schmidt procedure constructs orthonormal sets iteratively.

• Let U be a countable set of linearly independent vectors.

• we construct $\mathcal{U}$ such that $[\mathcal{U}] = [U]$.

Iteratively: $u_{n+1} = c_{n+1} \left(v_{n+1} - \sum_{k=1}^n (u_k, v_{n+1}) u_k \right) \neq 0$

$$[\{u_1, \dots, u_n\}] = [\{v_1, \dots, v_n\}]$$

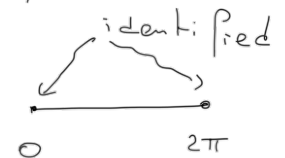
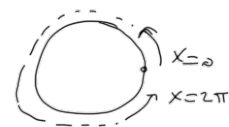
c_{n+1} chosen such that $\|u_{n+1}\| = 1$.

$$\text{So, } [\{u_1, \dots, u_{n+1}\}] = [\{v_1, \dots, v_{n+1}\}] \quad \square$$

• Fourier series (Chapter 7). Construction of an o.n.b.

• Let $f: \mathbb{R} \rightarrow \mathbb{C}$ be a 2π -periodic function $f(x+2\pi) = f(x) \quad \forall x \in \mathbb{R}$.

Then f is defined on "circle" $\mathbb{T}$



• In $\mathcal{C}(\mathbb{T})$, we define $\|f\| = \left(\int_0^{2\pi} |f(x)|^2 dx \right)^{1/2}$ and the completion of $\mathcal{C}(\mathbb{T})$ for this norm is $L^2(\mathbb{T})$.

• There is an inner product $\langle f, g \rangle = \int_0^{2\pi} \overline{f(x)} g(x) dx$ so that $(L^2(\mathbb{T}), \langle \cdot, \cdot \rangle)$ is a Hilbert space.

• We wish to construct an o.n.b. for that space.

• Basis: $e_n(x) = \frac{1}{\sqrt{2\pi}} e^{inx} = \frac{1}{\sqrt{2\pi}} (\cos nx + i \sin nx)$, $n \in \mathbb{Z}$.

$e_n(x) \in \mathcal{C}(\mathbb{T})$ and in fact $\mathcal{C}^\infty(\mathbb{T})$. $\langle e_n, e_m \rangle = \begin{cases} 1 & n=m \\ 0 & n \neq m \end{cases}$.

(check).

. Theorem. $\mathcal{U} = \{e_n(x), n \in \mathbb{Z}\}$ is o.n.b. for $L^2(\mathbb{T})$.

. Thus, $f(x) = \sum_{n \in \mathbb{Z}} \langle e_n, f \rangle e_n(x)$ for all $f \in L^2(\mathbb{T})$.

. $f_n := \langle e_n, f \rangle$ Fourier coefficients of $f(x)$.

. Proof. (1) Density. Lemma. Let $f_m \rightarrow f$ with $f_m = \sum_{n \in \mathbb{Z}} \langle e_n, f_m \rangle e_n$ $\textcircled{*}$.

Then $\langle e_n, f_m \rangle \xrightarrow{m \rightarrow \infty} \langle e_n, f \rangle$ AND $f = \sum_{n \in \mathbb{Z}} \langle e_n, f \rangle e_n$.

proof of Lemma. let us compute $\langle f, f_m \rangle = \langle f, \sum_n \langle e_n, f_m \rangle e_n \rangle = \sum_n \langle e_n, f_m \rangle \langle f, e_n \rangle$

From (a) $\Leftrightarrow$ (b) in earlier Thm, we want $\{\langle f, e_n \rangle = 0 \ \forall n \Rightarrow f = 0\}$.

But $\langle f, e_n \rangle = 0 \Rightarrow \langle f, f_m \rangle = 0 \ \forall m \Rightarrow \langle f, f \rangle = 0$ in limit $\Rightarrow f = 0$.

Note $f_m \rightarrow f$ weakly is sufficient to prove Lemma.

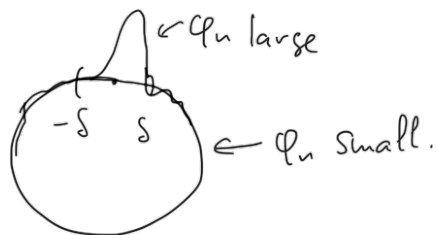
(2) Choice of dense family of functions f_m such that $\textcircled{*}$ holds.

. $\mathcal{C}(\mathbb{T})$ dense in $L^2(\mathbb{T})$. So we can choose f_m continuous.

. It remains to show $\textcircled{*}$ holds for continuous functions.

• f continuous. $f = \sum_n \langle e_n, f \rangle e_n$?

• Assume: $\varphi_n(x) \geq 0$ s.t. $\int_{-\pi}^{\pi} \varphi_n(x) dx = 1$ and $\forall \delta > 0, \int_{|x| > \delta} \varphi_n(x) dx \xrightarrow{n \rightarrow \infty} 0$.

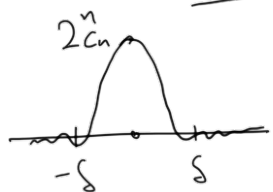


• Check that $\|\varphi_n * f - f\|_{L^2(\mathbb{T})} \rightarrow 0$ for $f \in C(\mathbb{T})$. (use ε - δ rule).

• Choice of $\varphi_n(x)$ now so that $\varphi_n * f$ satisfies $\oplus$ (use lemma again).

• We want $(\varphi_n * f)(x) = \int_{-\pi}^{\pi} \varphi_n(x-y) f(y) dy = \sum_{n \in \mathbb{Z}} \langle e_n, \varphi_n * f \rangle e_n$.

• Answer: $\varphi_n(x) = C_n (1 + \cos x)^n \geq 0$ with C_n such that $\int_{-\pi}^{\pi} \varphi_n(x) dx = 1$.



• verify $\int_{|x| > \delta} \varphi_n(x) dx \xrightarrow{n \rightarrow \infty} 0$ and $\varphi_n(x) = \sum_{k=-n}^n 2^{-n} \binom{2n}{n+k} e^{ikx}$.

• Thus, $(\varphi_n * f)(x) = \int_{-\pi}^{\pi} \varphi_n(x-y) f(y) dx = \sum_{k=-n}^n 2^{-n} C_n \binom{2n}{n+k} \left\{ \int_{-\pi}^{\pi} e^{-iky} f(y) dy \right\} e^{ikx}$
 $e^{ik(x-y)} = e^{ikx} e^{-iky}$! and $(\varphi_n * f)$ satisfies $\oplus$ □

